EXCEL CASE STUDY

Inventory segmentation and replenishment planning

Which products need closer review, and how sensitive are reorder points to changes in sales patterns?

Swamesh Lotlikar / UCI Online Retail / Jan–Nov 2011

Project scope and data source

461,414
Eligible transaction lines

3,715
SKUs analysed

£8.80m
Eligible product sales

Dataset

UCI Online Retail invoice lines from a UK-based retailer. Analysis covers eleven complete months, January through November 2011.

Planning question

Which SKUs deserve planning attention based on sales value and variability? What reorder points would different demand assumptions imply?

The 4.83 million eligible units sold serve as the demand proxy.

Transaction cleaning and reconciliation

Rule or disposition	Lines
Outside Jan–Nov 2011	68,006
Cancellation invoices	8,200
Non-positive quantity	1,236
Non-positive unit price	954
Outside product scope	2,098
Verified reversed sale	1
Eligible product sales	461,414
Source total	541,909

Product scope

Removed postage, fees, samples, vouchers and accounting adjustments from the planning population.

Verified reversal

Matched a 74,215-unit sale for SKU 23166 to an equal cancellation 16 minutes later. Excluding the positive leg removed £77,183.60 from eligible sales.

Status fields preserve the source rows and explain every exclusion.

Fields and calculations added in Excel

Workbook layer	Fields and calculations added
Transactions	Record and analysis status, reversal review, sales value, sales month
Monthly trends	Monthly sales value and quantity, filtered to eligible rows
SKU classification	Revenue share, cumulative share, monthly units, CV, ABC–XYZ and sparse-history flag
Replenishment	334 calendar days per example SKU, daily mean and spread, safety stock and ROP
Summary	Sales trend, ABC–XYZ matrix and scenario comparisons

Sales value = quantity × unit price. Sales month groups dated invoice lines.

ABC–XYZ classification rules

ABC measures sales importance

Rank SKUs by eligible sales value. Divide each SKU's sales value by the total, then accumulate those shares in rank order.

A ≈ first 80% of sales value
B ≈ next 15%
C ≈ remaining 5%

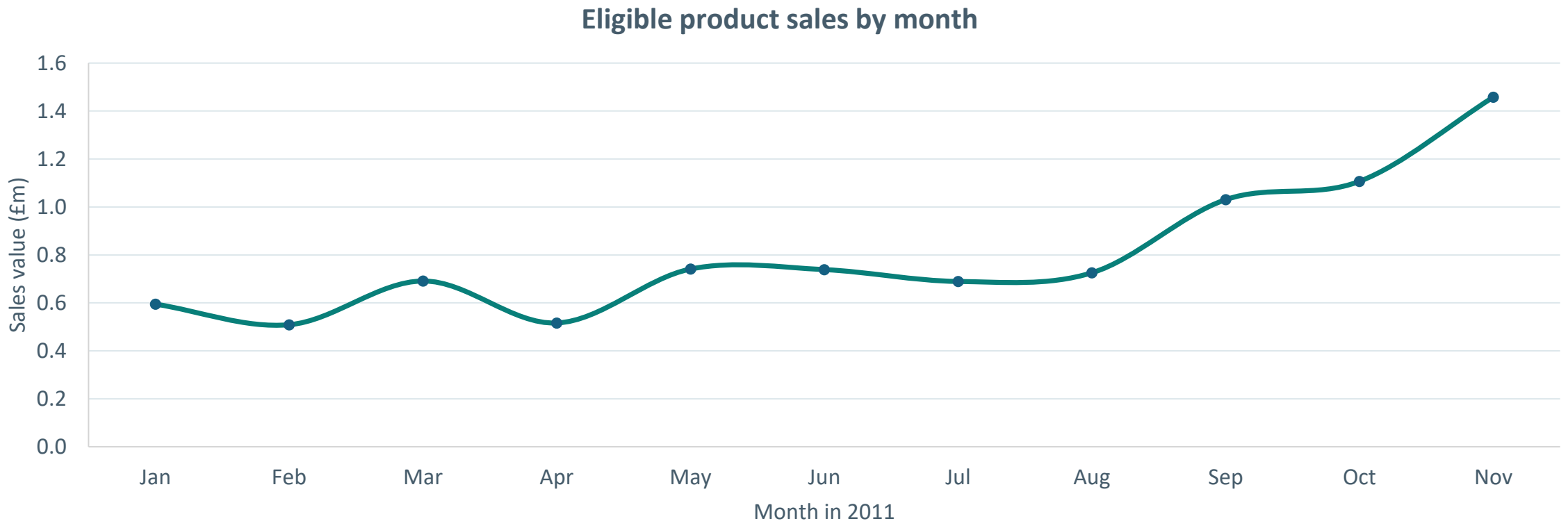
XYZ measures sales variability

Monthly CV = sample standard deviation of monthly units ÷ average monthly units. Include months with zero eligible sales.

X CV ≤ 0.50
Y 0.50 < CV ≤ 1.00
Z CV > 1.00

The workbook flags SKUs with sales in fewer than six months for review.

November led eligible product sales



Sales rose each month from August to November, reaching £1.46m. Use more years to test whether the peak recurs.

About 22% of SKUs account for 80% of sales value

ABC class	SKU count	Share of SKUs	Share of sales
A	816	21.97%	≈80%
B	941	25.33%	≈15%
C	1,958	52.71%	≈5%

SKU count	X	Y	Z	Total
A	214	329	273	816
B	119	345	477	941
C	20	374	1,564	1,958
Total	353	1,048	2,314	3,715

A-class planning focus

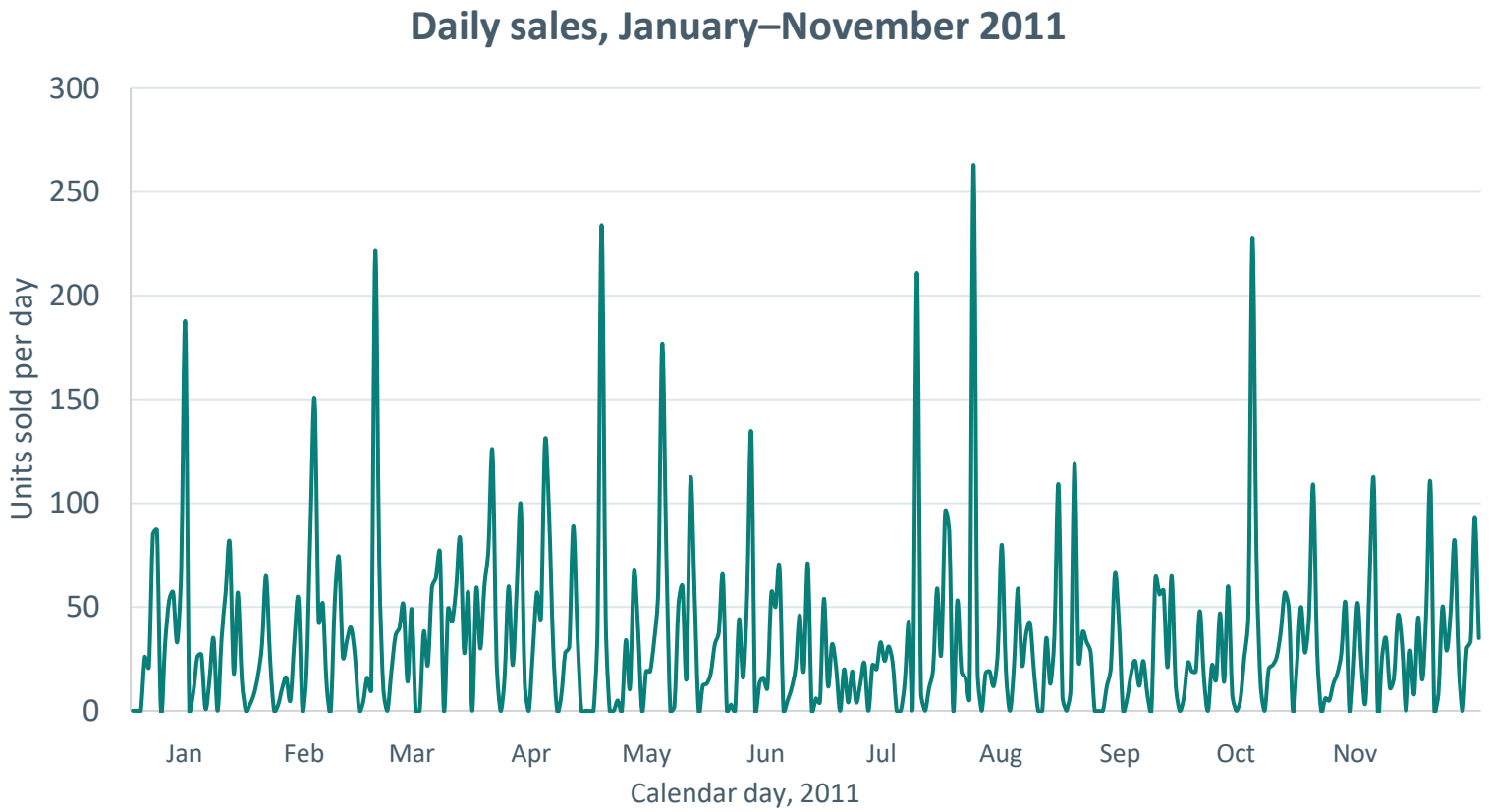
Review 214 AX items for routine replenishment. Investigate the demand patterns of 273 AZ items before setting buffers.

Safety stock and reorder point

Concept	Meaning and workbook calculation
Lead time	Days from placing an order to receipt. Assumed constant at 7 calendar days.
Cycle service level	Target probability of avoiding a stockout in a replenishment cycle. 95% gives z ≈ 1.645.
Expected lead-time demand	Average daily units × lead time in calendar days
Safety stock (SS)	Buffer against demand variability. Round up z × daily sample SD × √lead time.
Reorder point (ROP)	Trigger for replenishment. Round up expected lead-time demand + rounded SS.

Compare ROP with inventory position = on-hand + on-order − backorders.

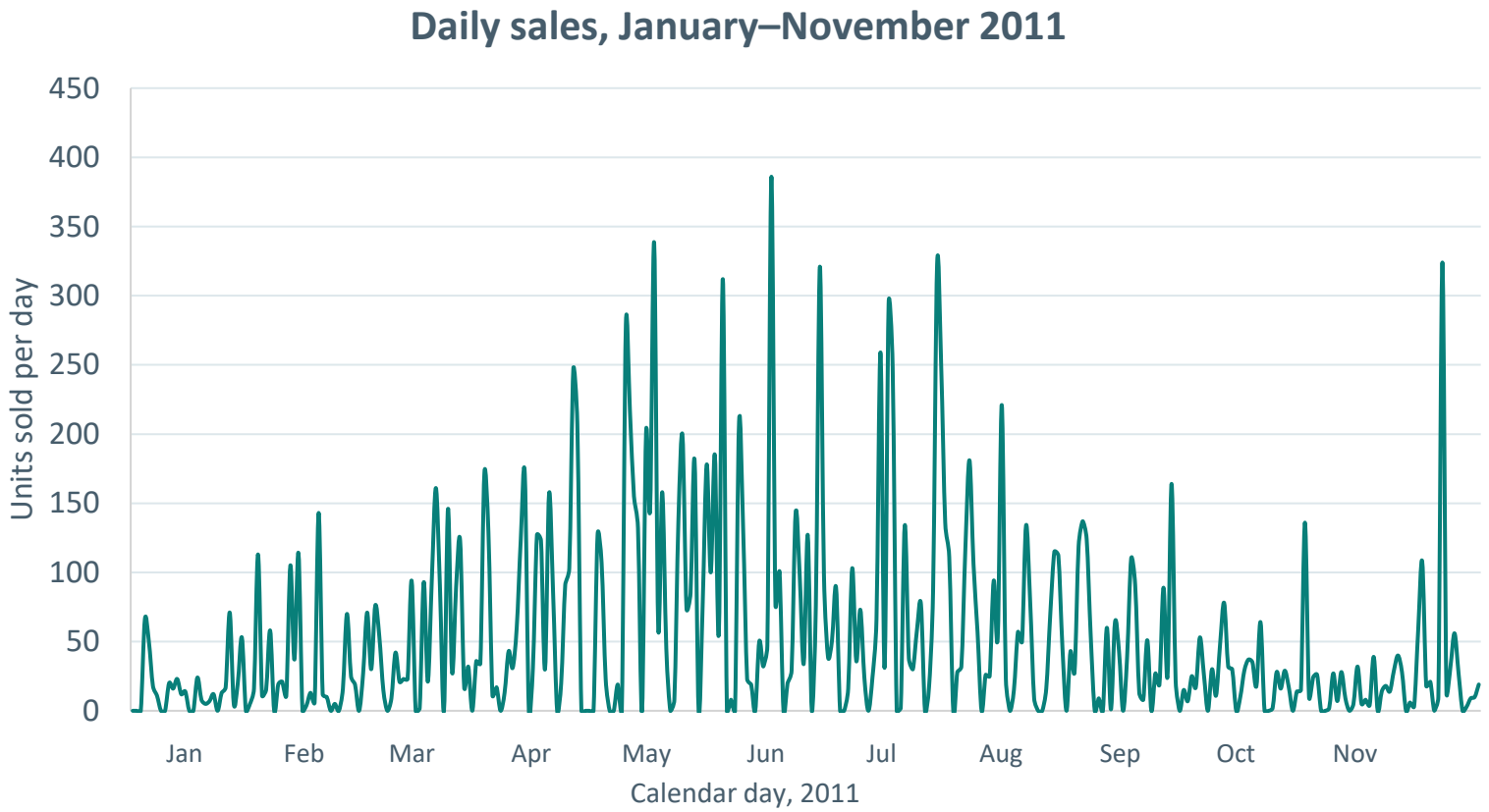
AX daily sales for SKU 22423



7 days / 95%	Units
Daily mean	33.94
Daily SD	39.49
Lead-time sales	237.58
Safety stock	172
Reorder point	410

Monthly sales are relatively steady. Use 410 units as a reference point when testing the model against actual lead times.

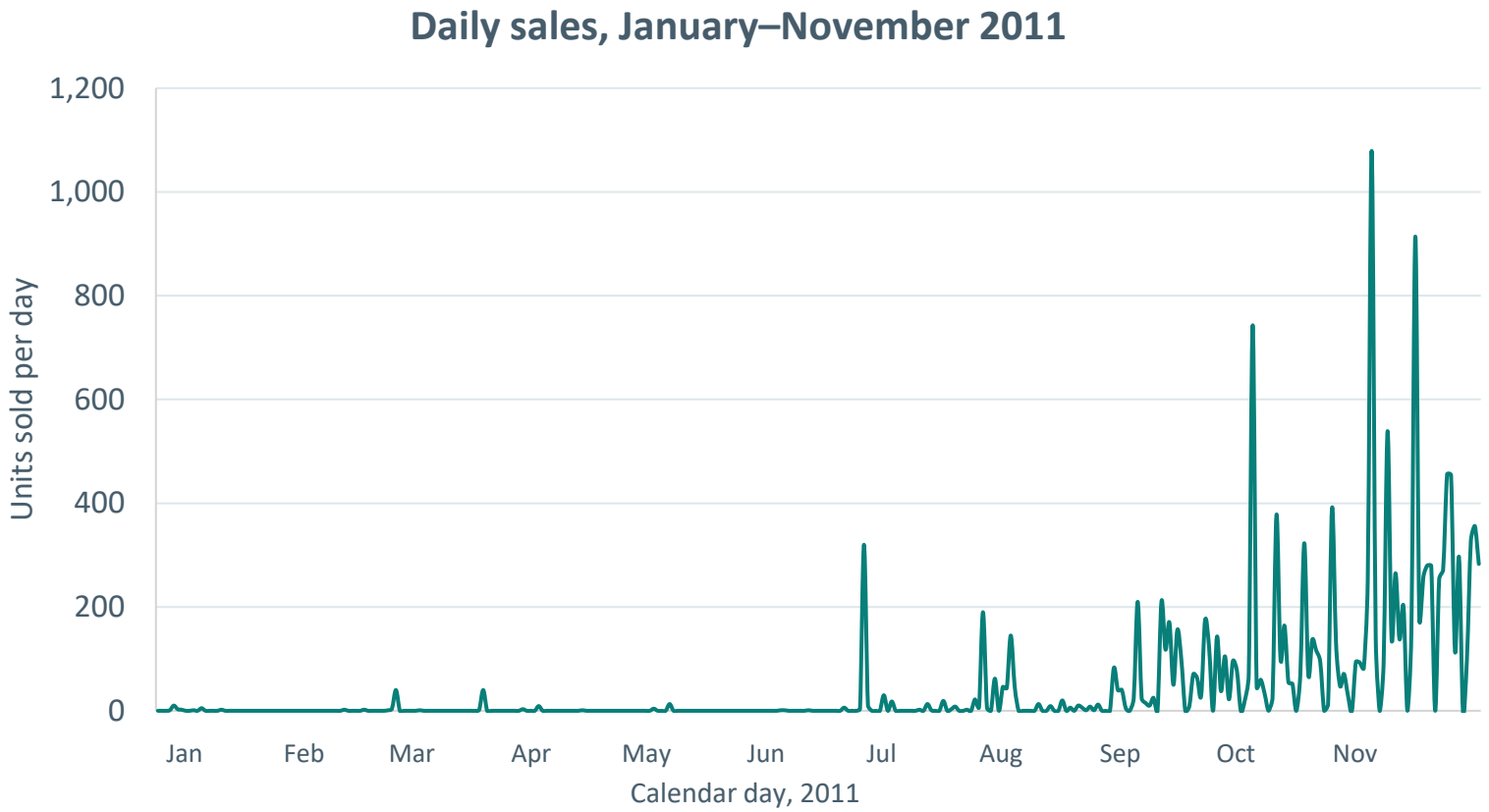
AY daily sales for SKU 47566



7 days / 95%	Units
Daily mean	53.66
Daily SD	69.57
Lead-time sales	375.61
Safety stock	303
Reorder point	679

Monthly sales peaked in May and generally fell afterward. Review the recent sales level before applying the full-period ROP.

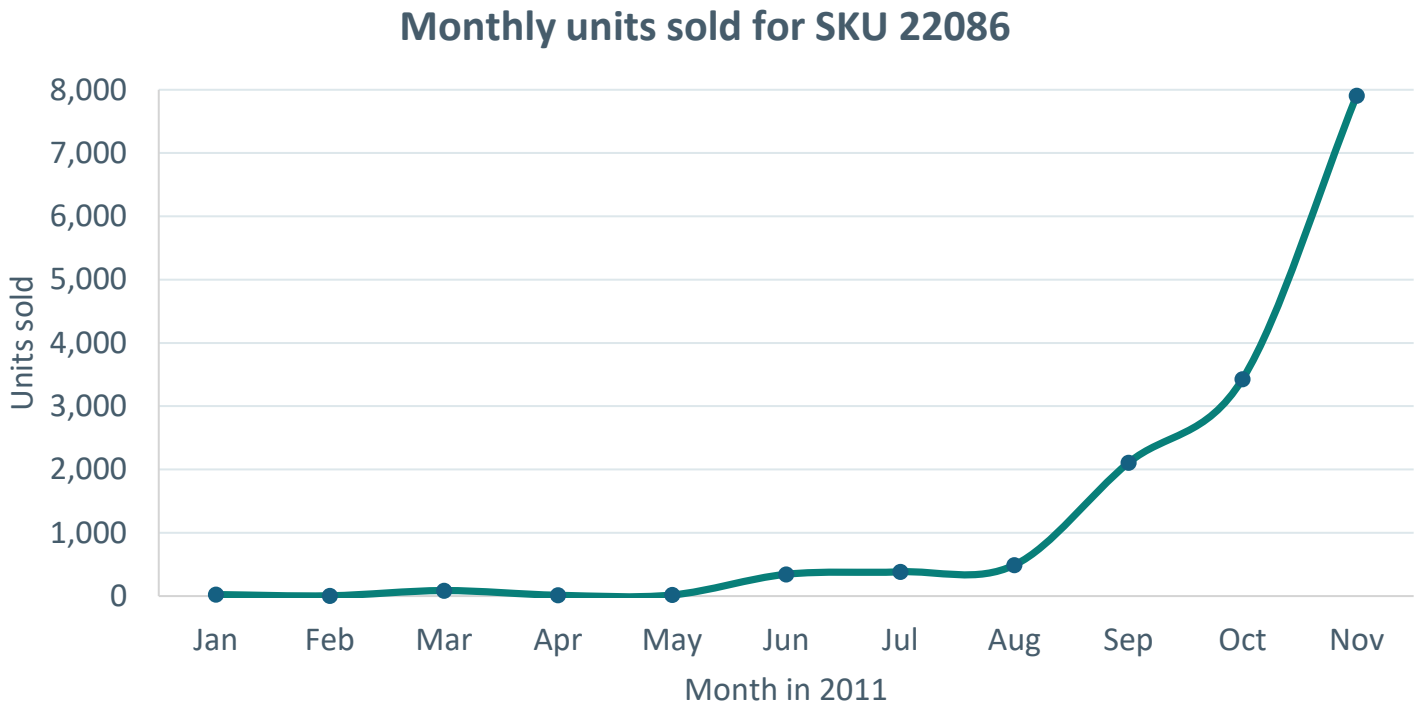
AZ daily sales for SKU 22086



7 days / 95%	Units
Daily mean	44.33
Daily SD	118.81
Lead-time sales	310.31
Safety stock	518
Reorder point	829

Sales rose sharply late in the year. The full-period ROP of 829 can mask the size of the recent uplift.

SKU 22086 sold 91% of its units in Sep–Nov



Observed period	Jan–Aug	Sep–Nov
Units sold	1,363	13,443
Calendar days	243	91
Daily mean	5.61	147.73
Daily SD	27.09	188.15

Monthly sales rose from 2,106 units in September to 7,908 in November. The recent daily spread includes that rise.

Two demand assumptions, two reorder points

Model input or result	Uplift fades	Recent level persists
Historical window used	Jan–Aug	Sep–Nov
Assumed daily units	5.61	147.73
Daily SD used	27.09	188.15
Expected lead-time units	39.26	1,034.08
Safety stock	118	819
Illustrative ROP	158	1,854

Both scenarios assume 7 calendar days and 95% cycle service. The recent case holds the Sep–Nov average flat. Its historical spread includes the sales ramp, which may overstate random variation.

Planning review priorities

Planning area	Review before implementation
AX	Check actual lead times and inventory position. Test the baseline ROP on a relatively stable item.
AY	Check whether the most recent sales level differs from the full-period average before setting the trigger.
AZ	Review promotions, bulk orders, stock availability and later sales to judge whether the uplift persists.
Service target	Compare the cost of carrying extra stock with the stockout risk. For 22423, moving from 95% to 99% adds 72 buffer units.